Views - *Assignment 11*

1) Create a view that shows all of the customers who have the highest ratings.

```
mysql> CREATE VIEW max_rating_customers AS SELECT cname FROM customers WHERE rating = (SELECT MAX
(rating) AS max_rating FROM customers);
Query OK, 0 rows affected (0.006 sec)

mysql> SELECT * FROM max_rating_customers;
+----------+
| cname    |
+----------+
| Grass    |
| Cisneros |
+----------+
2 rows in set (0.002 sec)
```

2) Create a view that shows the number of salespeople in each city.

```
mysql> CREATE VIEW no_salespeople_in_city AS SELECT city, COUNT(snum) FROM salespeople GROUP BY c
ity;
Query OK, 0 rows affected (0.005 sec)

mysql> SELECT * FROM no_salespeople_in_city;
+-----------+-------------+
| city      | COUNT(snum) |
+-----------+-------------+
| London    |           2 |
| San Jose  |           1 |
| Barcelona |           1 |
| New York  |           1 |
+-----------+-------------+
4 rows in set (0.004 sec)
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

```
mysql> CREATE VIEW sales_avg_sum AS SELECT s.sname, AVG(o.amt), SUM(o.amt) FROM orders o JOIN salespeo
ple s WHERE o.snum = s.snum GROUP BY s.sname;
Query OK, 0 rows affected (0.005 sec)

mysql> SELECT * FROM sales_avg_sum;
+---------+-------------+------------+
| sname   | AVG(o.amt)  | SUM(o.amt) |
+---------+-------------+------------+
| Peel    | 5127.356667 |   15382.07 |
| Serres  | 1848.716667 |    5546.15 |
| Motika  | 1900.100000 |    1900.10 |
| Rifkin  |  558.425000 |    1116.85 |
| Axelrod | 1713.230000 |    1713.23 |
+---------+-------------+------------+
5 rows in set (0.003 sec)
```

4) Create a view that shows each salesperson with multiple customers.

```
mysql> CREATE VIEW sales_multi_cus AS SELECT s.sname, c.snum, COUNT(*) FROM customers c JOIN salespeop
le s ON c.snum = s.snum GROUP BY c.snum, s.sname  HAVING COUNT(*) >1;
Query OK, 0 rows affected (0.004 sec)

mysql> SELECT * FROM sales_multi_cus;
+--------+------+----------+
| sname  | snum | COUNT(*) |
+--------+------+----------+
| Peel   | 1001 |        2 |
| Serres | 1002 |        2 |
+--------+------+----------+
2 rows in set (0.003 sec)

mysql>
```

Changing Values through Views.

5) Which of these views are updateable (will allow DML operations)?

#1 Create View Dailyorders -
as Select Distinct cnum, snum, onum, odate from Orders;
— > **NOT UPDATABLE**

#2 Create View Custotals
as Select cname, Sum (amt) Sum_Amt from Orders, Customers
where Orders.cnum=Customers.cnum
Group by cname;
—> **NOT UPDATABLE**

#3 Create view Thirdorders
as Select * from Dailyorders where
odate='1990-10-03';
—> **NOT UPDATABLE**

#4 Create view Nullcities
as Select snum, sname, city
from Salespeople
where city is NULL
OR sname BETWEEN 'A' and 'MZ';
—> **UPDATABLE**

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

```
mysql> CREATE VIEW Commissions AS SELECT snum, comm FROM salespeople WHERE comm BETWEEN 0.10 AND 0.20
WITH CHECK OPTION;
Query OK, 0 rows affected (0.003 sec)

mysql> SELECT * FROM Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.12 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.002 sec)

mysql> UPDATE Commissions SET comm = 0.25 WHERE id = 1001;
ERROR 1054 (42S22): Unknown column 'id' in 'where clause'
mysql> UPDATE Commissions SET comm = 0.25 WHERE snum  = 1001;
ERROR 1369 (HY000): CHECK OPTION failed 'sales.commissions'
mysql> UPDATE Commissions SET comm = 0.15 WHERE snum  = 1001;
Query OK, 1 row affected (0.003 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.15 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.001 sec)
```